

★ WEEK 8 · PART 3 · PRACTICE

The Midterm, Worked by Hand

Diagnose a Return Series; Trade a Pair Across Weeks 6-8

Two exam-style problems that use the whole first-half toolkit

Part 3's synthesis, with the arithmetic shown. We work two integrative problems by hand - reading a return series for mean, autocorrelation, and volatility clustering, then building a pairs trade from cointegration to a live Kalman hedge.

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PART 3 · APPLICATIONS

Two Integrative Problems, by Hand

Case A: diagnose one return series - stationarity, an ARMA signal, and an ARCH signal - and name the model. Case B: run a pair from cointegration to z-score to a Kalman hedge. Then a Claude Code lab.

What the First Half Gave You

Now we use all of it - with numbers

1

Model the mean, then the variance

Stationarity (W1-2), ARMA for the mean (W3-4), GARCH for volatility (W5). Reading a return series uses all three.

2

Go joint, then find equilibrium

VAR and Granger (W6), cointegration and the ECM (W7). This is where a tradable pair comes from.

3

Then make the hidden visible

State space and the Kalman filter (W8) track hidden states - like a hedge ratio that drifts. We tie it all into two problems.

Everything here is a calculator exercise: one return series, one pair, the whole first-half toolkit.

How These Cases Work

Exam-style problems, worked step by step

Each case has three steps

- Start from raw prices or returns.
- Compute each diagnostic by hand.
- Then name the model or the trade.

What you should watch

- How one series reveals two structures.
- How a pair goes from data to a live trade.
- Which week each number comes from.

Diagnose a Return Series

Weeks 1-2, 3, and 5 in one question

A classic exam question: here are eight returns - what model do they need? We compute three diagnostics by hand: the mean and variance (is it stationary?), the lag-1 autocorrelation (is there an ARMA signal?), and the autocorrelation of squared returns (is there volatility clustering?)

The three checks

- Stationarity: mean and variance (W1-2).
- Mean structure: lag-1 autocorrelation (W3).
- Variance structure: squared-return ACF (W5).

What we'll do by hand

- Compute the mean, variance, and sd.
- Compute the lag-1 autocorrelation.
- Compute the ACF of squared returns.

The Return Series

Eight days to diagnose

day	1	2	3	4	5	6	7	8
return (%)	+0.5	-0.4	+0.3	-0.6	-1.7	+1.9	-1.6	+1.8

Notice two things by eye: the signs tend to alternate (mean reversion), and the series is quiet early, then loud - a clustering hint.

Mean, Variance, Stationarity

The Week 1-2 check

$$\begin{aligned}\text{mean} &= \frac{1}{8} \sum r_t = 0.03 \approx 0 \\ \text{variance} &= \frac{1}{7} \sum (r_t - \bar{r})^2 = 1.88, \quad \text{sd} = 1.37 \\ &\Rightarrow \text{mean near zero, finite variance: looks stationary (W1-2)}\end{aligned}$$

Mean near zero and a finite, stable variance: nothing screams non-stationary, so the classical toolkit applies.

Lag-1 Autocorrelation

The Week 3 check - is the mean predictable?

$$\rho_1 = \frac{\sum_t (r_t - \bar{r})(r_{t-1} - \bar{r})}{\sum_t (r_t - \bar{r})^2} = -0.65 \Rightarrow \text{strong mean reversion: fit AR / ARMA (W3)}$$

The number

- The lag-1 autocorrelation is -0.65.
- Strongly negative - today reverses yesterday.
- The alternating signs made this visible.

The verdict

- This is real mean structure, not noise.
- An AR(1) with a negative coefficient fits.
- So the mean needs an ARMA model (W3).

Squared-Return Autocorrelation

The Week 5 check - is the variance predictable?

$$\rho_1^{(r^2)} = \frac{\sum_t (r_t^2 - \bar{r^2})(r_{t-1}^2 - \bar{r^2})}{\sum_t (r_t^2 - \bar{r^2})^2} = +0.62 \Rightarrow \text{volatility clusters: fit GARCH (W5)}$$

The number

- The squared-return ACF is +0.62.
- Positive - big moves follow big moves.
- That is volatility clustering.

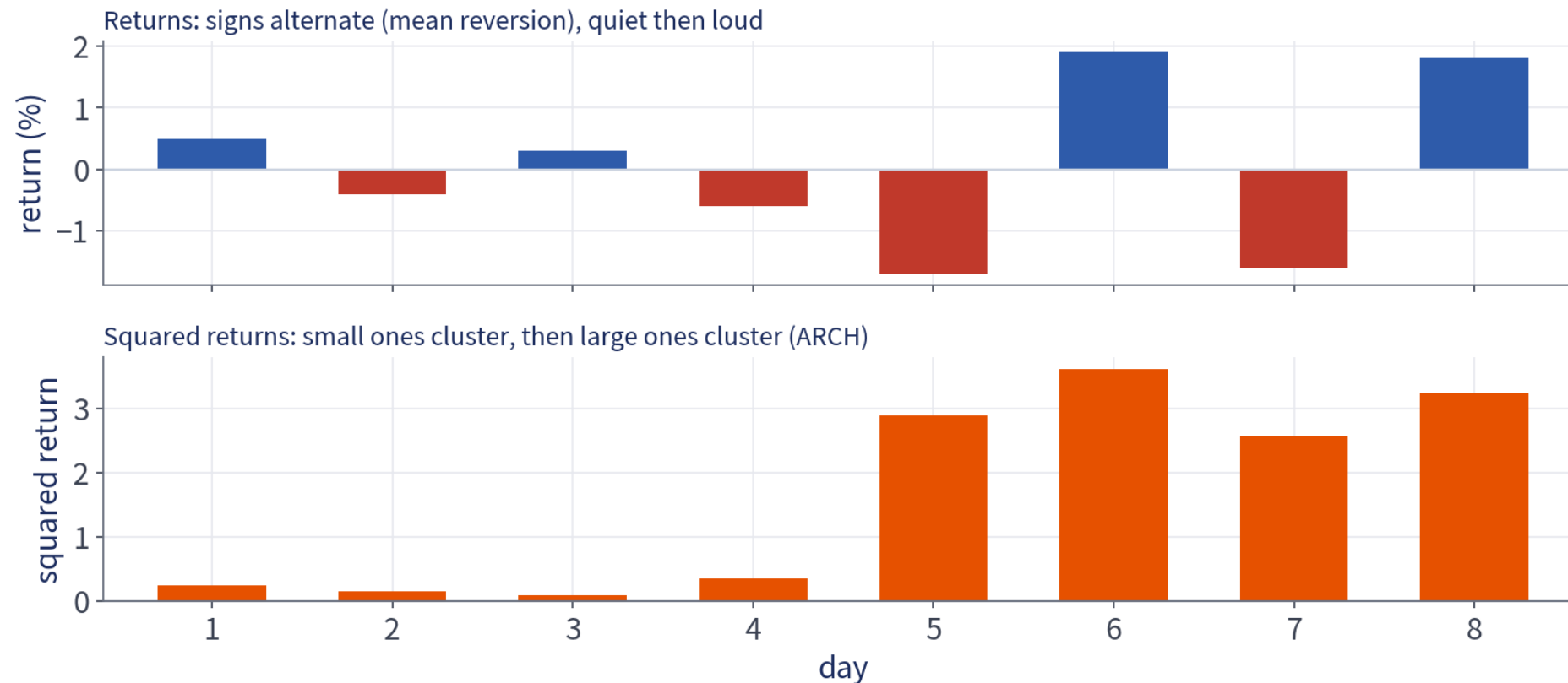
The verdict

- The variance is predictable, even if the sign isn't.
- This calls for an ARCH / GARCH model.
- So the variance needs GARCH (W5).

One Series, Two Structures

The returns (top) alternate in sign - the mean reversion an AR model captures. The squared returns (bottom) are quiet early and loud late - the volatility clustering GARCH captures. One series needs both a mean model and a variance model

Case A: one series, two structures - mean reversion and volatility clustering



The Full Diagnosis

What model does this series need?

What we found

- Stationary: mean ~ 0 , stable variance.
- Mean reverting: lag-1 autocorr -0.65.
- Clustered volatility: squared ACF +0.62.

What to fit

- An AR(1) or ARMA for the mean (W3).
- A GARCH for the variance (W5).
- Together: an ARMA-GARCH model.

A Pair, From Data to a Live Hedge

Weeks 6-8 in one question

The second integrative problem: two stocks that move together. We find the hedge ratio from cointegration (W7), turn the spread into a z-score entry (W1-3), then keep the hedge current with one Kalman update (W8). One pair walks through the whole second half of the first eight weeks

The three steps

- Cointegration: the hedge ratio and spread (W7).
- z-score: the entry signal (W1-3).
- Kalman: keep the hedge live (W8).

What we'll do by hand

- Compute beta from covariance and variance.
- Standardize today's spread into a z-score.
- Update beta with one new return pair.

The Hedge Ratio

Cointegration sets the pair (W7)

$$\beta = \frac{\text{Cov}(A,B)}{\text{Var}(B)} = \frac{0.074}{0.083} = \mathbf{0.89} \Rightarrow \text{spread} = A - 0.89 B \quad (\text{W7})$$

The calculation

- $\text{beta} = \text{Cov}(A,B) / \text{Var}(B) = 0.89$.
- It's the slope that best hedges A with B.
- The spread is A minus 0.89 times B.

Why it matters

- The spread is what you actually trade.
- A good hedge ratio makes it mean-reverting.
- This is the cointegration idea, applied.

The Entry Signal

A z-score triggers the trade (W1-3)

$$z = \frac{\text{spread}_{\text{today}} - \mu}{\sigma} = +1.8 \Rightarrow |z| > 1.5 : \text{spread is rich, SHORT it}$$

The calculation

- Standardize today's spread: $z = +1.8$.
- That's 1.8 standard deviations rich.
- Beyond the 1.5 sd band we set.

The action

- A rich spread should revert - so short it.
- Short A, long 0.89 of B, market-neutral.
- Exit as the spread returns toward its mean.

Keep the Hedge Live

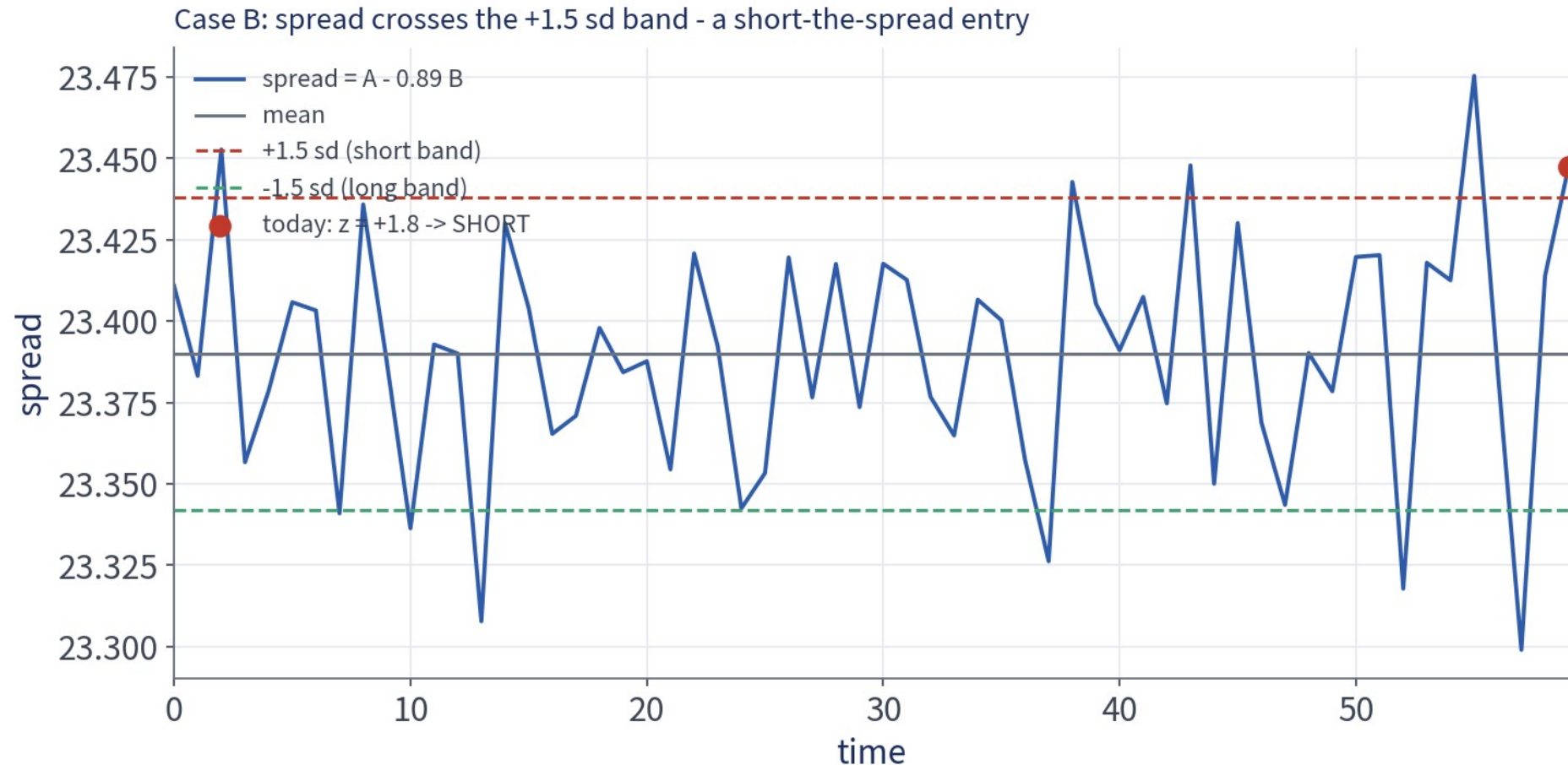
One Kalman update of beta (W8)

$$e = r_A - \beta r_B = 1.3 - 0.89(1.0) = 0.41$$
$$K = \frac{P r_B}{r_B^2 P + \sigma_\varepsilon^2} = \frac{0.021}{0.171} = 0.12, \quad \beta \leftarrow 0.89 + 0.12(0.41) = \mathbf{0.94} \text{ (W8)}$$

A new return pair with an innovation of 0.41 and a gain of 0.12 nudges the hedge ratio from 0.89 to 0.94 - the pair is kept in tune as it drifts.

The Spread and the Entry

The spread A minus $0.89 B$ oscillates around its mean. Today it crosses the $+1.5$ standard-deviation band at $z = +1.8$ - the short-the-spread entry. The Kalman filter keeps the 0.89 hedge ratio current as the relationship drifts



The Full Trade

Which week did each job

The trade

- Short the spread at $z = +1.8$.
- Hedge A with 0.94 of B, kept live.
- Exit as the spread reverts to its mean.

Which week did what

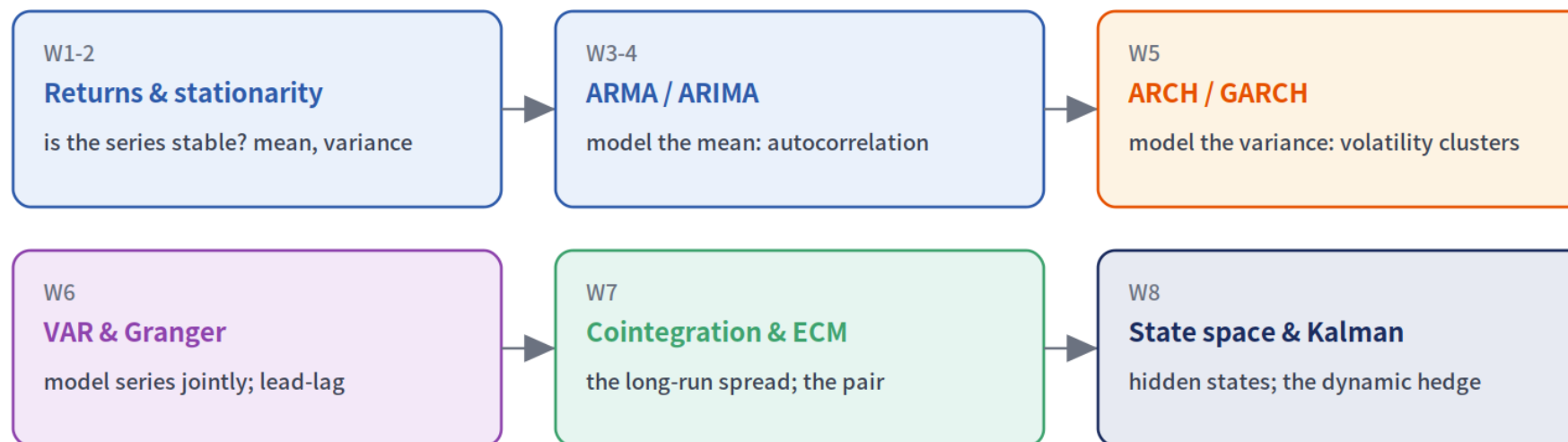
- W7 cointegration: the hedge ratio and spread.
- W1-3: the z-score entry signal.
- W8 Kalman: the drifting hedge, kept current.

The First-Half Toolbox, at a Glance

Both cases drew on the same eight-week arc: stationarity, ARMA, GARCH, VAR, cointegration, and state space. Read a series to pick a mean and variance model; build a pair to trade it - with the pairs thread tying every week together

The First-Half Toolbox, at a Glance

Eight weeks, one connected story: model the mean, find it is unpredictable, model the variance, go joint, then hidden.



The pairs-trading thread ties them together

Cointegration finds the pair (W7); a z-score triggers entry (W1-3); GARCH sizes it (W5); Kalman keeps the hedge current (W8).

How to Attack a Midterm Question

Diagnose, then decide

The habit

- Always check stationarity first.
- Separate the mean from the variance.
- Say which week's tool each finding calls for.

The payoff

- A messy series becomes a clear model choice.
- A pair of prices becomes a defined trade.
- You show you can connect the weeks.

Reproduce Both Diagnostics

Direct Claude Code to check your arithmetic

GOAL Verify the return-series diagnosis and the pairs pipeline, then run them on real data.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - diagnose:

- * take the 8 returns; print mean, variance, and sd;
- * print the lag-1 autocorrelation of returns;
- * print the lag-1 autocorrelation of squared returns;
- * name the model (ARMA for mean, GARCH for var).

CASE B - the pair:

- * from the price table, compute $\beta = \text{Cov}/\text{Var}$;
- * form the spread and today's z-score;
- * do one Kalman update of β .

Match every printed number to the slides.

Check against the slides

- mean ~ 0 , sd 1.37.
- return autocorr -0.65.
- squared autocorr +0.62.
- beta 0.89 $\rightarrow$ 0.94, $z = 1.8$.

Deliverable: `part3_practice.py` whose printed numbers match both worked diagnoses.

Three Questions Before the Exam

If you can answer these, you're ready

Q1 What does a negative return autocorrelation suggest?

**Mean reversion - fit
ARMA**

Q2 What does a positive squared-return ACF suggest?

**Volatility clustering -
fit GARCH**

Q3 Which weeks build a pairs trade?

**Cointegration (W7),
z-score (W1-3),
Kalman (W8)**

All three come from the return-series diagnosis and the pairs pipeline.

Three Pitfalls to Avoid

Where midterm answers slip

Common mistakes

- "Returns are unpredictable, so skip the ACF."
- "One model handles mean and variance."
- "A fixed hedge ratio is fine forever."

The reality

- Check both return and squared-return ACF.
- The mean and the variance need separate models.
- A drifting hedge needs a Kalman update.

Two Problems to Remember

Part 3 applications in one screen

1

Diagnose a series in three checks

Mean ~ 0 with stable variance (stationary), a lag-1 autocorrelation of -0.65 (mean reversion, fit ARMA), and a squared-return ACF of +0.62 (volatility clustering, fit GARCH). One series, two structures.

2

Build a pair from data to a live hedge

Cointegration gave $\beta = 0.89$ and the spread; today's z-score of +1.8 triggered a short; one Kalman update kept the hedge current at 0.94. Weeks 7, 1-3, and 8, in one trade.

3

Connect the weeks

The midterm rewards showing how the tools fit together - stationarity, ARMA, GARCH, cointegration, and Kalman - with the pairs thread running through them all.

Next — Week 9 and the second half: the Kalman filter deepens, and new state-space models arrive.